Dr. Akhilesh Das Gupta Institute of Professional Studies

Department of Artificial Intelligence & Machine Learning

Institute of Professional Studies

Cloud Computing

(OAE316P)

SUBMITTED BY: SUBMITTED TO:

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PRACTICAL 1

AIM:

Install VirtualBox/VMware workstation 45 5 install a C compiler in the virtual machine and execute a sample program.

THEORY:

Virtualization software like VirtualBox and VMware workstation allows users to create and manage virtual machines on their host operating system. These virtual machines can be used to run multiple operating systems simultaneously, providing flexibility and isolation for various tasks, such as software development, testing, and experimentation.

C is a powerful programming language commonly used for system programming, software development, and more. To compile and execute C programs, a C compiler needs to be installed within the operating system environment.

PROCEDURE:

1. Install VirtualBox or VMware Workstation:

    a. Download the appropriate installer for VirtualBox or VMware Workstation from their respective official website.
    b. Follow the installation instructions provided by the installer for your operating system.
    c. Launch the virtualization software once the installation is complete.

2. Create a New Virtual Machine:

    a. Open VirtualBox or VMware Workstation.
    b. Click on "New" to create a new virtual machine.
    c. Follow the wizard to set up the virtual machine, specifying parameters such as memory, storage, and operating system installation ISO.

3. Install an Operating System:

    a. Start the virtual machine.
    b. Follow the prompts to install an operating system within the virtual machine. For this practical, you can choose a lightweight Linux distribution such as Ubuntu Server or CentOS.

4. Install a C Compiler:

a. Once the operating system is installed and running within the virtual machine, open a terminal or command prompt.
b. Install a C compiler such as GCC (GNU Compiler Collection) using the package manager of the operating system.
   i. For Ubuntu/Debian: **sudo apt-get install build-essential**
   ii. For CentOS/RHEL: **sudo yum groupinstall "Development Tools"**

5. Write and Compile a Sample C Program:

   a. Using a text editor (e.g., Vim, Nano), write a simple C program, such as a "Hello, World!" program.
   b. Save the program with a **.c** extension (e.g., **hello.c**).
   c. Compile the C program using the installed compiler:

```
gcc -o hello hello.c
```

- If there are no syntax errors, the compilation will generate an executable file named **hello**.

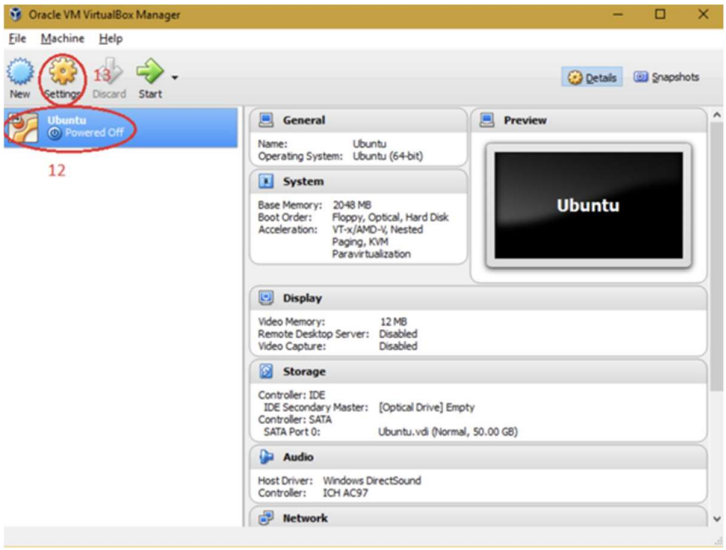
Execute the Sample C Program:

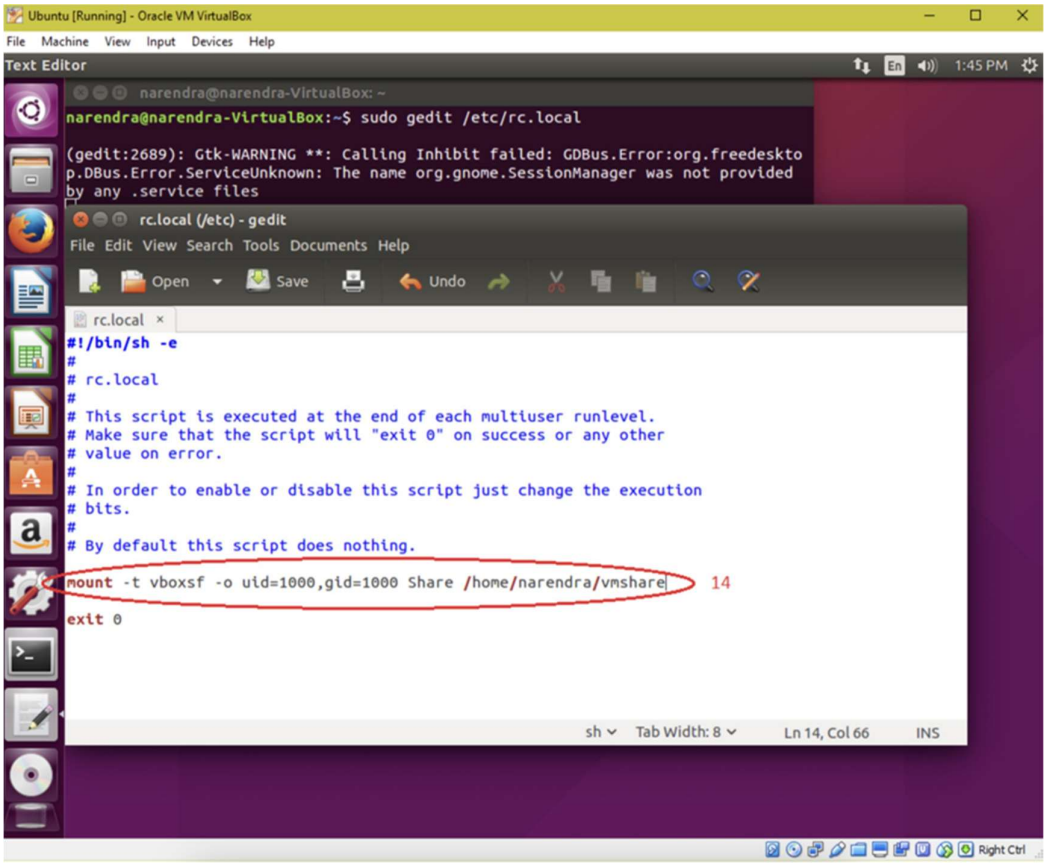
- Run the compiled C program:

```
./hello
```

You should see the output of the program, which should display "Hello, World!".

RESULT:


Oracle VM VirtualBox Manager
File Machine Help
New
Settings
13
Discard
Start
Details
Snapshots
Ubuntu
Powered Off
12
General
Name: Ubuntu
Operating System: Ubuntu (64-bit)
System
Base Memory: 2048 MB
Boot Order: Floppy, Optical, Hard Disk
Acceleration: VT-x/AMD-V, Nested Paging, KVM Paravirtualization
Preview
Ubuntu
Display
Video Memory: 12 MB
Remote Desktop Server: Disabled
Video Capture: Disabled
Storage
Controller: IDE
IDE Secondary Master: [Optical Drive] Empty
Controller: SATA
SATA Port 0: Ubuntu.vdi (Normal, 50.00 GB)
Audio
Host Driver: Windows DirectSound
Controller: ICH AC97
Network


Ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Text Editor
1:45 PM
narendra@narendra-VirtualBox: ~
narendra@narendra-VirtualBox:~$ sudo gedit /etc/rc.local
(gedit:2689): Gtk-WARNING **: Calling Inhibit failed: GDBus.Error:org.freedesktop.DBus.Error.ServiceUnknown: The name org.gnome.SessionManager was not provided by any .service files
rc.local (/etc) - gedit
File Edit View Search Tools Documents Help
Open
Save
Undo
rc.local
#!/bin/sh -e
#
rc.local
#
This script is executed at the end of each multiuser runlevel.
Make sure that the script will "exit 0" on success or any other
value on error.
#
In order to enable or disable this script just change the execution
bits.
#
By default this script does nothing.
mount -t vboxsf -o uid=1000,gid=1000 Share /home/narendra/vmshare
14
exit 0
sh
Tab Width: 8
Ln 14, Col 66
INS
Right Ctrl

PRACTICAL 2

AIM:

Create type 2 virtualization in VMware. Allocate memory and storage space as per requirement. Install guest OS on that VMware.

THEORY:

Type 2 virtualization, also known as hosted virtualization, involves running a hypervisor on top of a host operating system. VMware Workstation is a popular Type 2 hypervisor that allows users to create and manage multiple virtual machines on their desktop or laptop computers. Each virtual machine can run its own guest operating system, providing isolation and flexibility for various tasks.

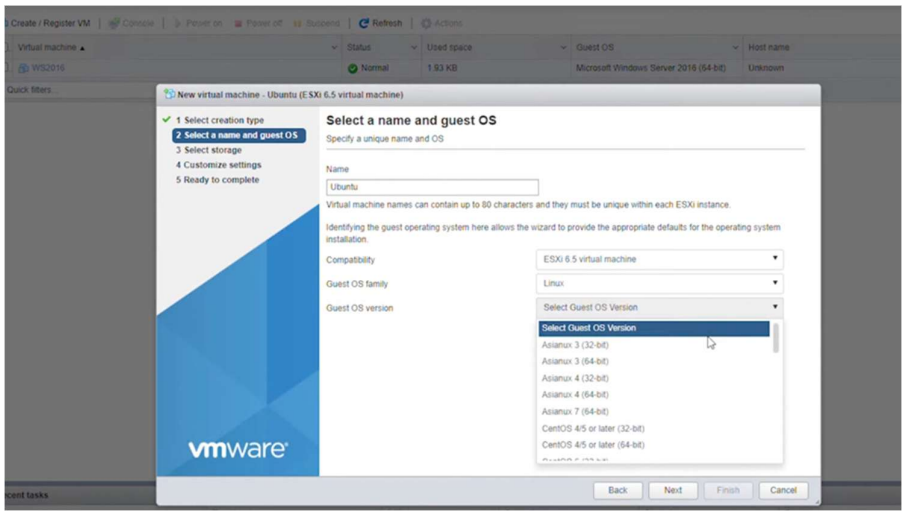
PROCEDURE:

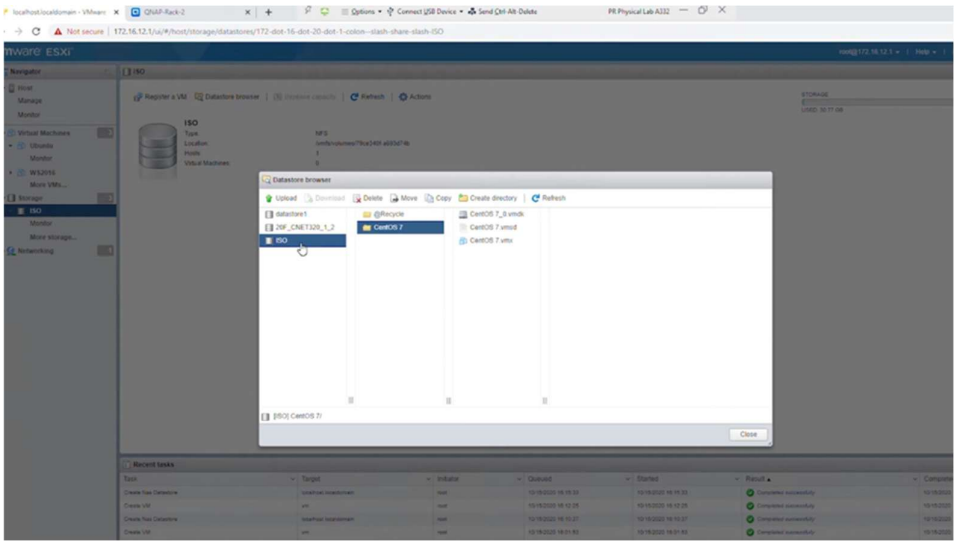
1. Install VMware Workstation:
    a. Download the VMware Workstation installer from the official website.
    b. Follow the installation instructions provided by the installer for your operating system.
    c. Launch VMware Workstation once the installation is complete.
2. Create a New Virtual Machine:
    a. Open VMware Workstation.
    b. Click on "File" > "New Virtual Machine" to start the New Virtual Machine Wizard.
    c. Choose "Typical" configuration and click "Next".
3. Select Guest Operating System:
    a. Choose the operating system you want to install as the guest OS. For example, if you want to install Ubuntu, select "Linux" and then choose the Ubuntu version from the dropdown menu.
    b. Click "Next".
4. Allocate Memory:
    a. Specify the amount of memory (RAM) to allocate to the virtual machine. Allocate memory based on the requirements of the guest OS and the applications you plan to run within it.
    b. Click "Next".
5. Allocate Disk Space:

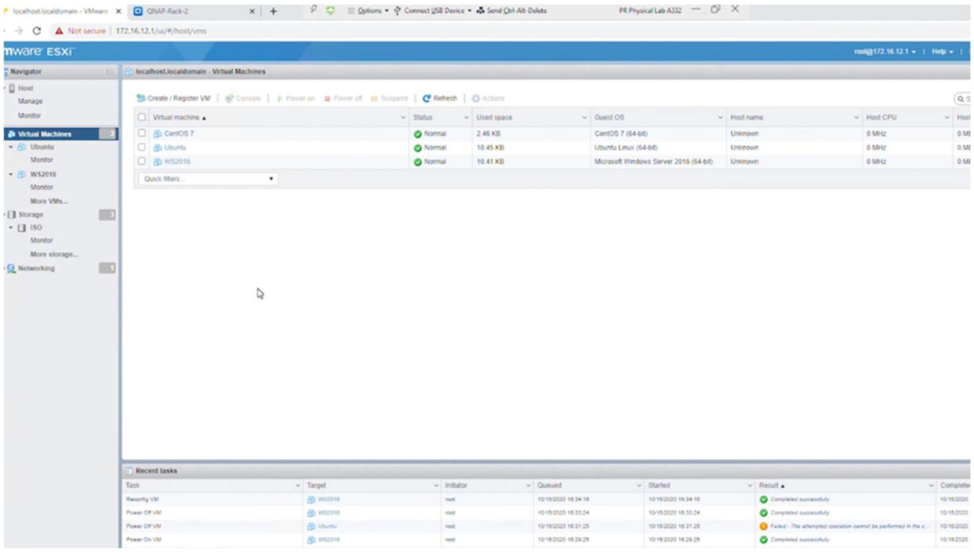
a. Choose whether to create a new virtual disk or use an existing one. For this practical, let's create a new virtual disk.
b. Specify the disk capacity. Allocate disk space based on the requirements of the guest OS and the applications you plan to install.
c. Choose whether to store the virtual disk as a single file or split into multiple files.
d. Click "Next".

6. Customize Hardware (Optional):
   a. You can customize hardware settings such as CPU cores, network adapter, display resolution, etc., according to your requirements.
   b. Once customization is complete, click "Finish" to create the virtual machine.
7. Install Guest Operating System:
   a. Start the virtual machine by clicking "Power on this virtual machine".
   b. Follow the prompts to install the guest operating system within the virtual machine. Insert the installation media (e.g., ISO file) if required.
   c. Complete the installation process by following the on-screen instructions provided by the guest OS installer.

RESULT:







PRACTICAL 3

AIM:

Adding a new virtual disk to a virtual machine. Convert basic disc to dynamic disc and vice versa

a. Shrink and extend virtual disk
b. Create, manage, configure and schedule snapshots
c. Create spanned, mirrored and striped volume
d. Create raid 5 volume

THEORY:

In virtual machine environments, disk management tasks play a crucial role in optimizing storage resources, enhancing performance, and ensuring data integrity. The following tasks are commonly performed:

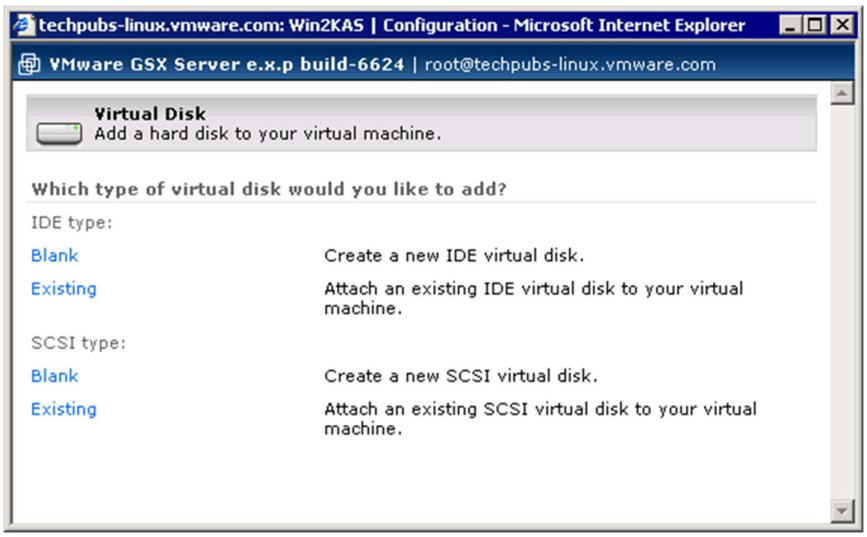
1) Adding a New Virtual Disk: This involves attaching an additional virtual disk to a virtual machine to increase storage capacity.
2) Converting Basic Disk to Dynamic Disk and Vice Versa: Basic disks are traditional disk storage types, while dynamic disks offer more advanced features like spanning volumes across multiple disks.
3) Shrinking and Extending Virtual Disks: Resizing virtual disks allows for adjusting storage capacity based on changing requirements.
4) Managing Snapshots: Snapshots capture the state of a virtual machine at a specific point in time, enabling easy rollback to previous configurations.
5) Creating Different Volume Types:
   a) Spanned Volume: Combines free space from multiple disks into a single logical volume.
   b) Mirrored Volume: Provides redundancy by duplicating data across multiple disks for fault tolerance.
   c) Striped Volume: Distributes data evenly across multiple disks to improve performance.
   d) RAID 5 Volume: Uses disk striping with parity to achieve both performance and data redundancy.

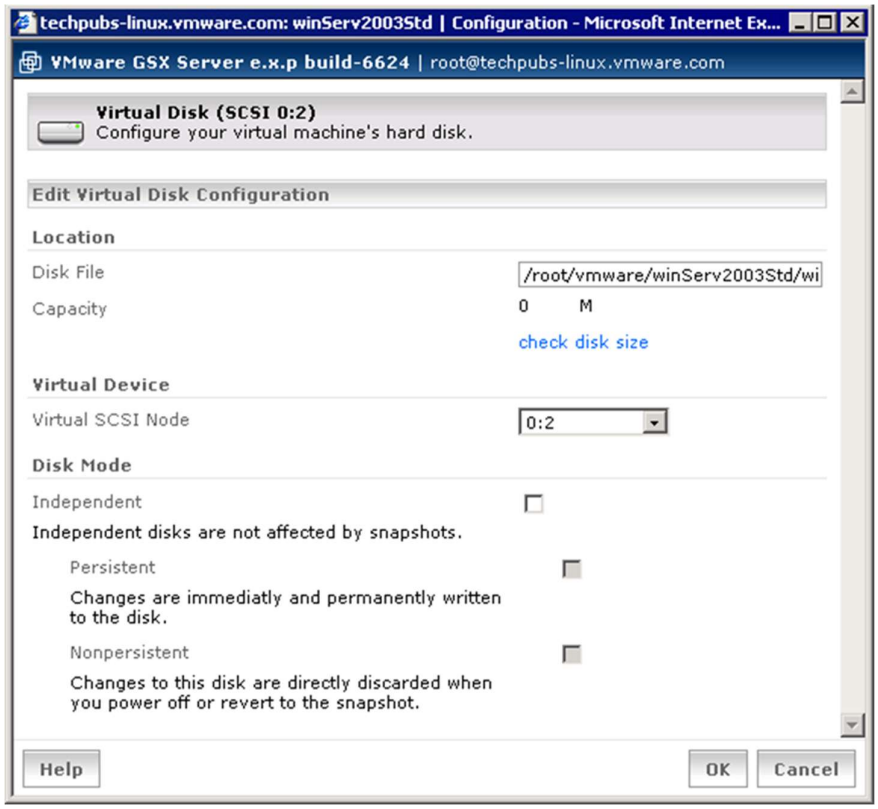
PROCEDURE:

1. Adding a New Virtual Disk:

   a. Open the virtual machine settings in your virtualization software (e.g., VMware Workstation).
   b. Add a new virtual disk and specify its size and type (e.g., SCSI, SATA).
   c. Save the settings and start the virtual machine.
2. Converting Basic Disk to Dynamic Disk and Vice Versa:
   a. In Windows, open Disk Management.
   b. Right-click on the disk you want to convert and choose "Convert to Dynamic Disk" or "Convert to Basic Disk" as appropriate.
   c. Follow the prompts to complete the conversion process.
3. Shrinking and Extending Virtual Disks:
   a. In your virtualization software, select the virtual machine and navigate to its settings.
   b. Choose the virtual disk you want to resize.
   c. Use the options to shrink or extend the disk size as needed.
4. Managing Snapshots:
   a. Take a snapshot of the virtual machine at its current state.
   b. Perform changes or updates to the virtual machine.
   c. If needed, revert to the snapshot to restore the virtual machine to its previous state.
5. Creating Different Volume Types:
   a. In Windows, open Disk Management.
   b. Right-click on "Disk Management" and choose the appropriate option to create a new volume (spanned, mirrored, striped, RAID 5).
   c. Follow the wizard to configure the volume settings, including disk selection and allocation size.

RESULT:





PRACTICAL 4

AIM:

Sharing and data transfer between the virtual machines.

THEORY:

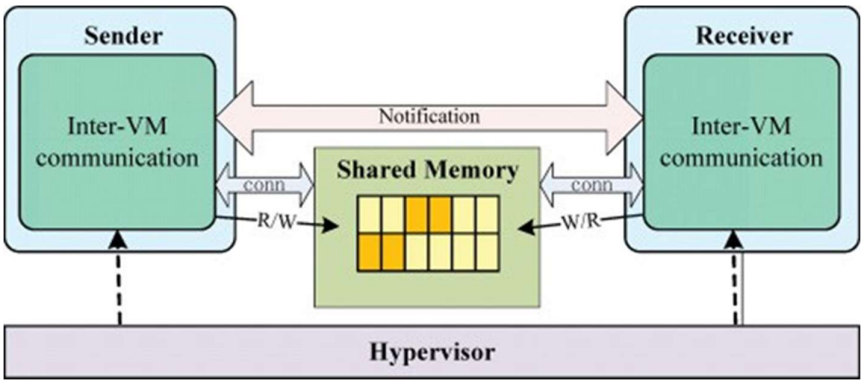
In virtualized environments, sharing and transferring data between virtual machines are common requirements for collaboration, data exchange, and resource utilization. Various methods can be employed to achieve this, including shared folders, network file sharing protocols, and direct data transfer mechanisms.

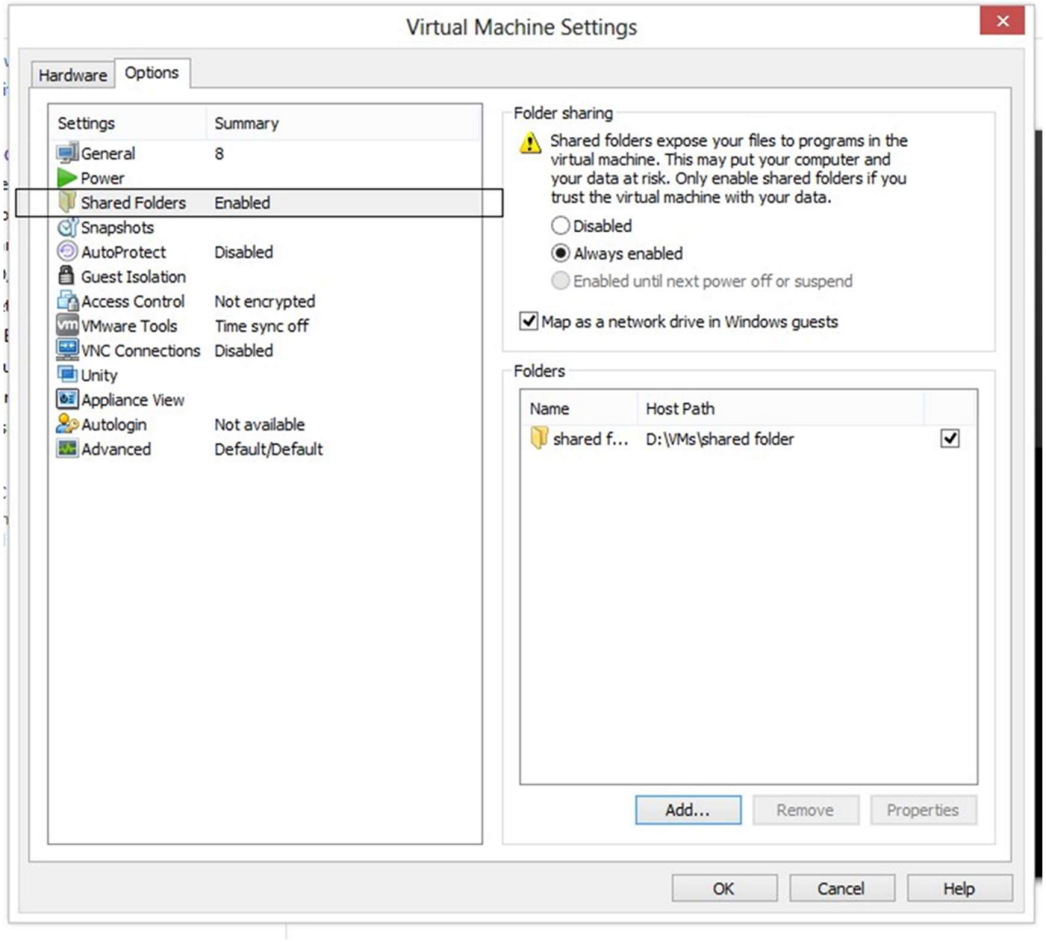
PROCEDURE:

1. Shared Folders:
    a. Many virtualization platforms offer the capability to create shared folders that can be accessed by multiple virtual machines.
    b. In VMware Workstation, for example, you can set up shared folders by navigating to VM settings > Options > Shared Folders.
    c. Enable shared folders and specify the host folder(s) that you want to share with the virtual machine.
    d. Access the shared folder from within the virtual machine's operating system to transfer files between the host and guest systems.
2. Network File Sharing:
    a. Set up network file sharing protocols such as SMB (Server Message Block) or NFS (Network File System) to share files between virtual machines.

   b. Configure file sharing settings on the host system and within the virtual machines to allow access to shared directories.
   c. Access shared files using file explorer or terminal commands from within the virtual machines.
3. Direct Data Transfer:
   a. Use direct data transfer methods such as SCP (Secure Copy Protocol) or SFTP (SSH File Transfer Protocol) to transfer files between virtual machines.
   b. Enable SSH (Secure Shell) on the virtual machines and use SCP or SFTP commands to transfer files securely over the network.
   c. Authenticate using appropriate credentials (e.g., username and password) to establish a secure connection for data transfer.
4. Shared Clipboard:
   a. Some virtualization software allows for shared clipboards between the host system and virtual machines.
   b. Enable shared clipboard functionality in the virtualization software settings.
   c. Copy text or files on the host system and paste them directly into applications running within the virtual machines, and vice versa.
5. Shared Network:
   a. Ensure that all virtual machines are connected to the same virtual network within the virtualization software.
   b. Assign IP addresses to each virtual machine within the same subnet to facilitate network communication.
   c. Use standard networking protocols (e.g., TCP/IP) to transfer data between virtual machines over the network.

RESULT:





PRACTICAL 5

AIM:

Create type 2 virtualization on ESXi 6.5 server.

THEORY:

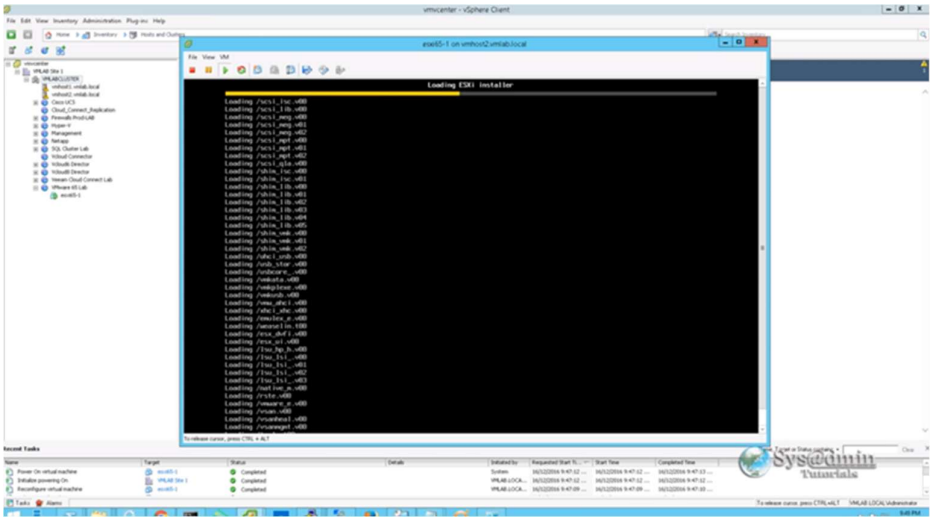
Type 2 virtualization involves running a hypervisor within a host operating system. In this case, we'll use VMware Workstation as the Type 2 hypervisor, installed on macOS. VMware Workstation allows you to create and manage virtual machines on your local machine, which can then be connected to and run on your ESXi 6.5 server.

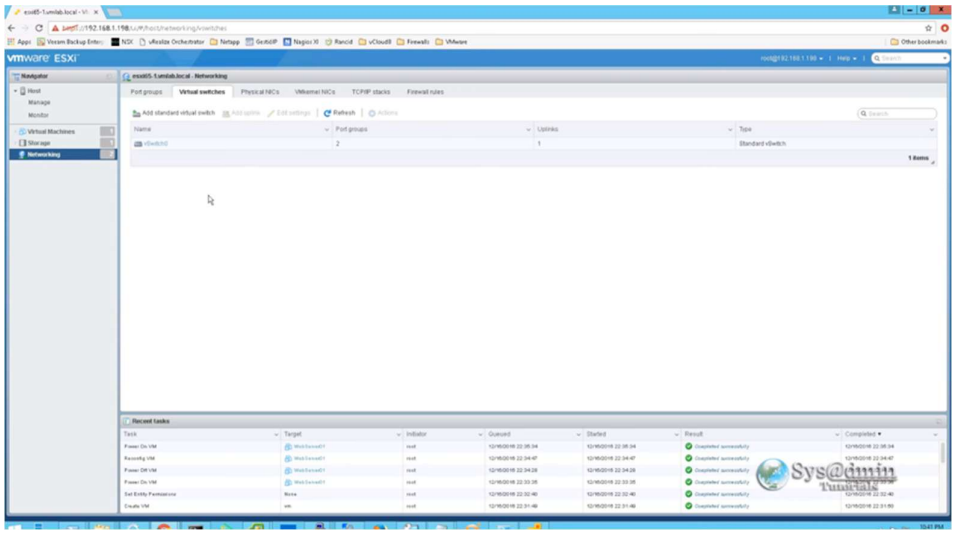
PROCEDURE:

1. Install VMware Workstation on macOS:
    a. Download the VMware Workstation for macOS installer from the VMware website.

   b. Double-click the installer file and follow the on-screen instructions to install VMware Workstation on your macOS system.
2. Connect to Your ESXi Server:
   a. Launch VMware Workstation on your macOS.
   b. Click on **File > New Virtual Machine**.
   c. Select **New Remote Server**.
   d. Enter the IP address or hostname of your ESXi server.
   e. Enter your ESXi server username and password when prompted.
   f. Click **Connect** to establish a connection to your ESXi server.
3. Create a Virtual Machine on ESXi:
   a. Once connected to your ESXi server, click on **Create a New Virtual Machine**.
   b. Follow the wizard to create a new virtual machine. Specify the operating system, virtual hardware configuration, and allocate resources like CPU cores, memory, and storage.
   c. Customize any additional settings as needed.
4. Install Guest Operating System:
   a. After creating the virtual machine, power it on.
   b. Follow the installation procedure to install the guest operating system of your choice. This involves attaching an ISO file containing the installation media to the virtual machine and going through the installation process.
5. Configure Networking:
   a. Configure networking for your virtual machine to ensure it can communicate with other systems on your network or the internet. You can choose between bridged, NAT, or host-only networking modes depending on your requirements.
6. Manage Virtual Machines:
   a. You can now manage your virtual machines directly from VMware Workstation on your macOS. This includes powering them on/off, taking snapshots, and configuring virtual hardware settings.
7. Additional Considerations:
   a. Ensure that your macOS system meets the hardware requirements for running VMware Workstation and hosting virtual machines.
   b. Keep your virtual machine files and configurations backed up regularly to prevent data loss.

RESULT:


vmvcenter - vSphere Client
esxi65-1 on vmhost2.vmlab.local
Loading ESXi installer
Sys@dmin
Tutorials


vmware ESXi
Sys@dmin
Tutorials

PRACTICAL 6

AIM:

Create a VLAN in cisco packet tracer.

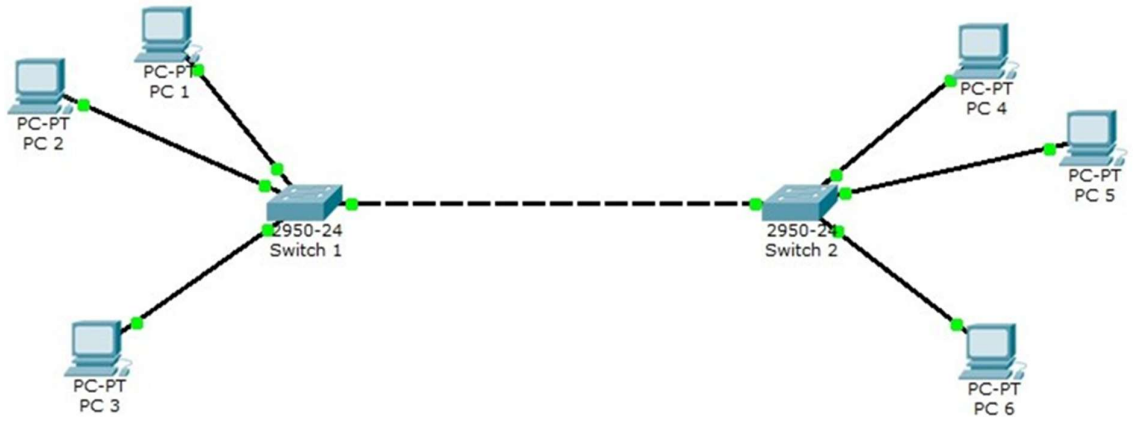
THEORY:

A VLAN (Virtual Local Area Network) is a logical grouping of devices within the same broadcast domain, even if they are not physically connected on the same network switch. VLANs are useful for segmenting a network to improve performance, security, and management.

PROCEDURE:

1. Launch Cisco Packet Tracer:
    a. Open Cisco Packet Tracer on your computer.
2. Create a Network Topology:
    a. Drag and drop the devices you want to include in your network topology. Typically, this includes switches, routers, and PCs.
3. Configure Switches:
    a. Select a switch in your topology and click on it to open its configuration.
    b. Enter the CLI (Command Line Interface) by clicking on the **CLI** tab.
    c. Enter privileged EXEC mode by typing **enable** and pressing **Enter**.
    d. Enter global configuration mode by typing **configure terminal** and pressing Enter.
4. Create VLANs:
    a. To create a VLAN, use the command **vlan <vlan_id>** followed by a VLAN number. For example, **vlan 10**.
    b. Repeat this step to create additional VLANs as needed.
5. Assign Ports to VLANs:
    a. Navigate to the interfaces you want to assign to VLANs. This could be individual ports or port ranges.
    b. Enter interface configuration mode by typing **interface <interface_id>** and pressing Enter. For example, **interface GigabitEthernet0/1**.
    c. Assign the interface to a VLAN using the command **switchport access vlan <vlan_id>**. For example, **switchport access vlan 10**.
6. Configure Trunk Ports (Optional):
    a. If you want to allow traffic from multiple VLANs to pass through a single switch port (trunking), configure the port as a trunk port.
    b. Enter interface configuration mode for the desired port.
    c. Use the command **switchport mode trunk** to configure the port as a trunk port.
7. Verify VLAN Configuration:
    a. After configuring VLANs and assigning ports, verify the configuration using the **show vlan** command to display the VLAN information and **show interfaces <interface_id> switchport** to verify port assignments.
8. Test Connectivity:
    a. Connect devices to the switch ports assigned to VLANs and test connectivity between devices within the same VLAN.
9. Save Configuration:
    a. Once you are satisfied with the VLAN configuration, save the configuration of the switches to ensure that changes are preserved. Use the **write** or **copy running-config startup-config** command.

RESULT:


PC-PT
PC 1
PC-PT
PC 2
PC-PT
PC 3
2950-24
Switch 1
2950-24
Switch 2
PC-PT
PC 4
PC-PT
PC 5
PC-PT
PC 6

PRACTICAL 7

AIM:

Find procedure to set up the one node Hadoop cluster.

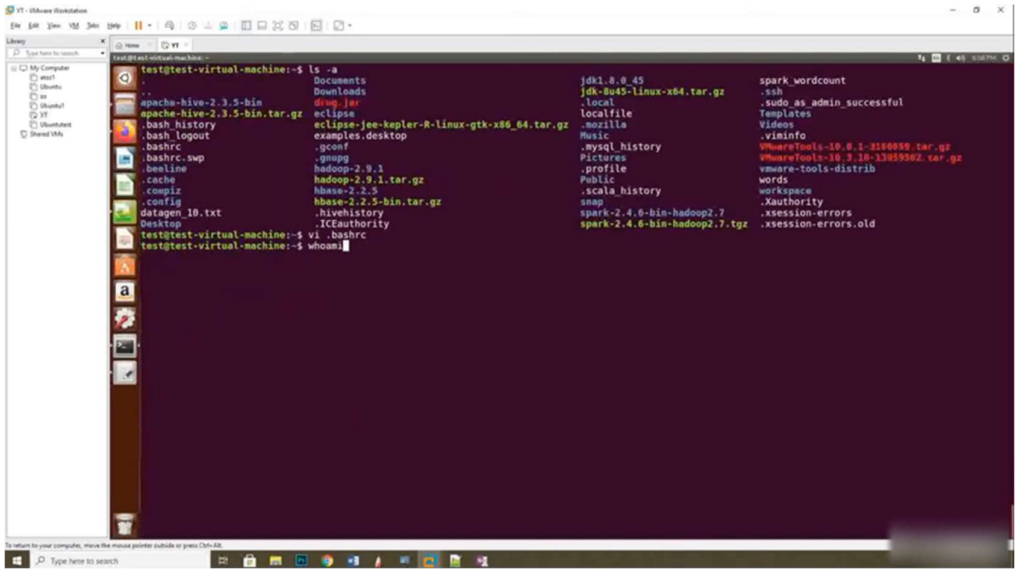
THEORY:

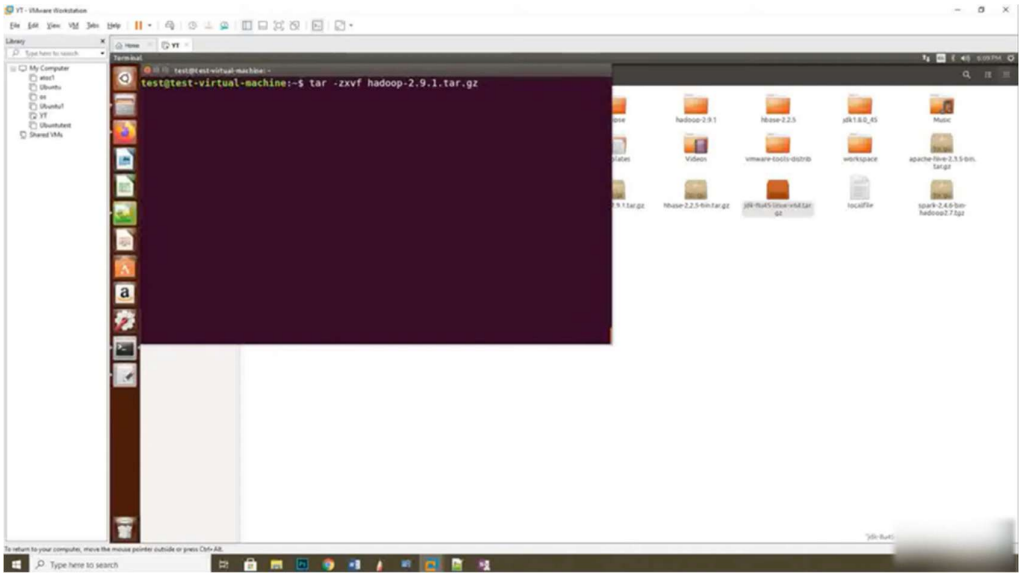
A one-node Hadoop cluster, also known as a pseudo-distributed Hadoop cluster, is a simplified setup where all Hadoop daemons run on a single machine. This configuration is suitable for learning, development, and testing purposes. In a one-node setup, Hadoop's core components, such as the Hadoop Distributed File System (HDFS) and Yet Another Resource Negotiator (YARN), are installed and configured to run on the same machine.

PROCEDURE:

1. Install Prerequisites:
    a. Ensure that your system meets the prerequisites for running Hadoop, including having Java Development Kit (JDK) installed.
2. Download and Extract Hadoop:
    a. Visit the Apache Hadoop website and download the desired version of Hadoop.
    b. Extract the downloaded Hadoop archive to a directory on your machine where you want to install Hadoop.
3. Configure Environment Variables:
    a. Set the **HADOOP_HOME** environment variable to point to the directory where Hadoop is installed.
    b. Update the **PATH** environment variable to include the bin and sbin directories within the Hadoop installation directory.
4. Configure Hadoop:
    a. Navigate to the **etc/hadoop** directory within your Hadoop installation directory.
    b. Modify the configuration files (**core-site.xml, hdfs-site.xml, yarn-site.xml**, etc.) to specify cluster settings such as the default filesystem, replication factor, and resource management configurations.
5. Format the Hadoop Filesystem:
    a. Use the **hdfs namenode -format** command to initialize the Hadoop filesystem. This command creates the initial filesystem metadata and data structures.
6. Start Hadoop Services:
    a. Start the Hadoop daemons using the provided scripts (**start-dfs.sh** and **start-yarn.sh**). These scripts launch the HDFS and YARN services respectively.
7. Verify Hadoop Installation:
    a. Access the Hadoop web interfaces to verify that the cluster is up and running. You can typically access the Namenode web interface at **http://localhost:9870** and the ResourceManager web interface at **http://localhost:8088**.
8. Run Hadoop Jobs:
    a. Use the Hadoop command-line tools (**hadoop fs, hadoop jar**, etc.) to interact with the Hadoop cluster and run MapReduce jobs or other Hadoop applications.
9. Experiment and Learn:
    a. Explore various Hadoop components, run sample jobs, and experiment with different configurations to learn more about Hadoop's capabilities and ecosystem.

RESULT:


test@test-virtual-machine:~$ ls -a
.
..
apache-hive-2.3.5-bin
apache-hive-2.3.5-bin.tar.gz
.bash_history
.bash_logout
.bashrc
.bashrc.swp
.beeline
.cache
.compiz
.config
datagen_10.txt
Desktop
Documents
Downloads
drug.jar
eclipse
eclipse-jee-kepler-R-linux-gtk-x86_64.tar.gz
examples.desktop
.gconf
.gnupg
hadoop-2.9.1
hadoop-2.9.1.tar.gz
hbase-2.2.5
hbase-2.2.5-bin.tar.gz
.hivehistory
.ICEauthority
jdk1.8.0_45
jdk-8u45-linux-x64.tar.gz
.local
localfile
.mozilla
Music
.mysql_history
Pictures
.profile
Public
.scala_history
snap
spark-2.4.6-bin-hadoop2.7
spark-2.4.6-bin-hadoop2.7.tgz
spark_wordcount
.ssh
.sudo_as_admin_successful
Templates
Videos
.viminfo
VMwareTools-10.0.1-3160050.tar.gz
VMwareTools-10.3.10-13959562.tar.gz
vmware-tools-distrib
words
workspace
.Xauthority
.xsession-errors
.xsession-errors.old
test@test-virtual-machine:~$ vi .bashrc
test@test-virtual-machine:~$ whoami


test@test-virtual-machine:~$ tar -zxvf hadoop-2.9.1.tar.gz

PRACTICAL 8

AIM:

Implement cloud security controls such as encryption, access management, and network security using cloud-native services.

THEORY:

Cloud security controls are essential for protecting data, applications, and infrastructure in cloud environments. These controls typically encompass encryption, access management, and network security.

Encryption:

- Encryption ensures that data is protected both at rest and in transit. At rest, encryption secures data stored in cloud databases, storage services, and backups. In transit, encryption safeguards data as it moves between users and cloud services, preventing interception and unauthorized access.

Access Management:

- Access management involves controlling and monitoring who can access cloud resources and what actions they can perform. Identity and Access Management (IAM) solutions are used to manage user identities, roles, and permissions. By implementing the principle of least privilege, organizations can ensure that users only have access to the resources necessary to perform their tasks.

Network Security:

- Network security focuses on protecting cloud networks from unauthorized access, data breaches, and other cyber threats. This involves segmenting the network using virtual private clouds (VPCs) or virtual networks (VNets), implementing firewalls and security groups to control traffic, and deploying distributed denial of service (DDoS) protection to mitigate attacks.

PROCEDURE:

1. Encryption:
    a. Data Encryption at Rest:
        i. Choose a cloud-native encryption service such as AWS Key Management Service (KMS), Google Cloud Key Management Service (KMS), or Azure Key Vault.
        ii. Enable encryption for storage services such as Amazon S3, Google Cloud Storage, or Azure Blob Storage.
        iii. Implement encryption for databases using features like Transparent Data Encryption (TDE) for SQL databases.
    b. Data Encryption in Transit:
        i. Use Transport Layer Security (TLS) or Secure Sockets Layer (SSL) protocols to encrypt data transmitted over the network.
        ii. Implement HTTPS for web applications to encrypt data between clients and servers.
        iii. Manage SSL/TLS certificates using cloud-native services like AWS Certificate Manager or Google Cloud SSL Certificates.

2. Access Management:
    a. Identity and Access Management (IAM):

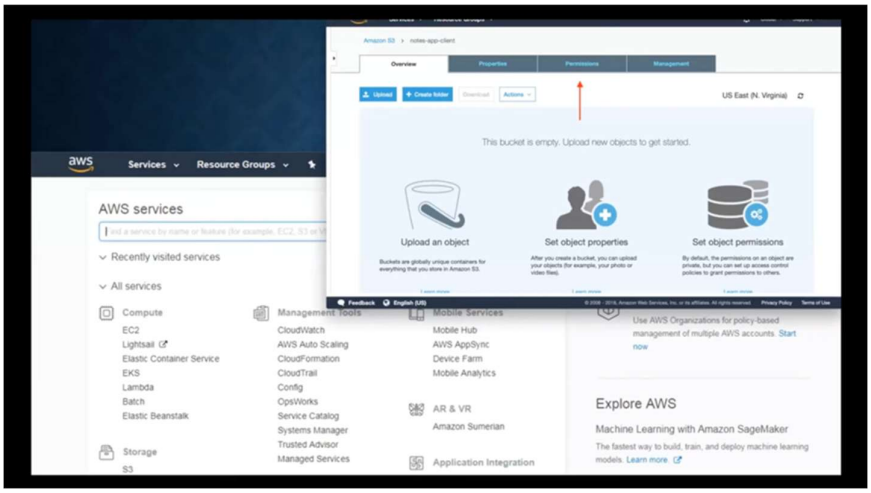
i. Utilize cloud provider IAM services such as AWS IAM, Google Cloud IAM, or Azure Active Directory (Azure AD) to manage user identities and access permissions.
ii. Implement role-based access control (RBAC) and assign permissions based on roles and responsibilities.
iii. Enable multi-factor authentication (MFA) for user access to enhance security.

b. Resource Policies:
i. Implement resource-level access controls using policies.
ii. Use AWS IAM policies, Google Cloud IAM policies, or Azure Resource Manager policies to control access to specific resources.

3. Network Security:
a. Virtual Private Cloud (VPC) or Virtual Network (VNet):
i. Segment the cloud network using VPCs (in AWS) or VNets (in Azure) to isolate resources and control traffic flow.
ii. Configure network access control lists (NACLs) or security groups to restrict inbound and outbound traffic at the subnet or instance level.
b. Firewalls and Security Groups:
i. Set up cloud provider firewalls or security groups to filter traffic based on IP addresses, ports, and protocols.
ii. Limit access to only necessary ports and protocols to reduce the attack surface.
c. DDoS Protection:
i. Enable DDoS protection services provided by cloud providers such as AWS Shield, Google Cloud Armor, or Azure DDoS Protection to mitigate DDoS attacks.

4. Continuous Monitoring and Compliance:
a. Implement logging and monitoring solutions provided by cloud providers to monitor cloud infrastructure and detect security incidents.
b. Enable auditing and logging for cloud services to track access and configuration changes.
c. Regularly review security configurations to ensure compliance with industry standards and regulations.

RESULT:



PRACTICAL 9

AIM:

Launch a virtual instance on a cloud platform and install a simple web server.

THEORY:

Cloud platforms provide virtual compute resources that can be provisioned on demand. This allows for flexible deployment of applications, such as web servers, without the need for physical hardware management. Virtual instances offer scalability, isolation, and cost efficiency based on usage.

PROCEDURE:

1. Create an Account:

    a. Access a cloud platform's management console and create or log in to an account.

2. Launch a Virtual Instance:

    a. Navigate to the compute or virtual machine section in the console.
    b. Select "Create Instance" or similar option.
    c. Choose the operating system (e.g., a Linux distribution).
    d. Specify instance details like machine type, region, and network settings.
    e. Configure security rules to allow HTTP (port 80) and SSH (port 22) access.
    f. Launch the instance.

3. Connect to the Instance:

    a. Use SSH to connect to the public IP of the instance using provided credentials.

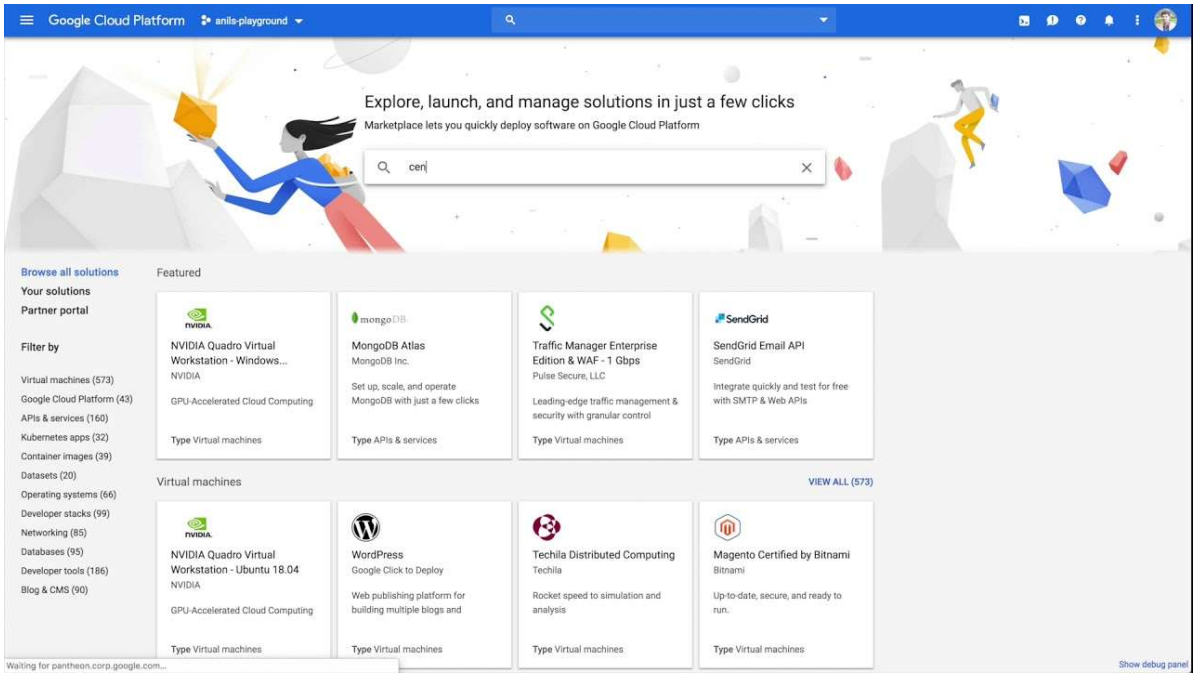
4. Install Web Server:

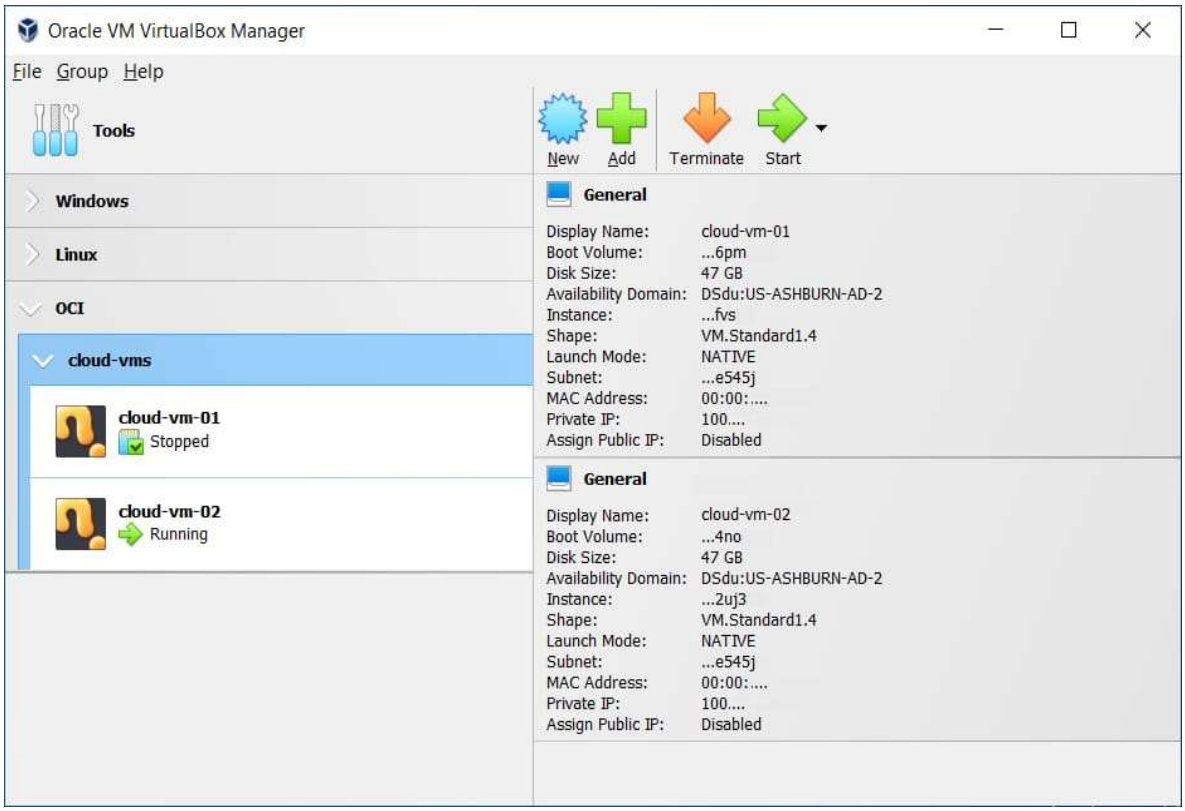
    a. Update packages: sudo apt update -y (for Debian-based systems).
    b. Install a web server package: sudo apt install apache2 -y.
    c. Start the service: sudo systemctl start apache2.
    d. Enable on boot: sudo systemctl enable apache2.

5. Verify:

    a. Access the public IP in a browser to see the default web page.

RESULT:





PRACTICAL 10

AIM:

Create a storage bucket on a cloud platform and upload a file.

THEORY:

Cloud storage services provide durable, scalable object storage for data. Buckets act as containers for files, enabling easy upload, retrieval, and management of data with features like versioning and access controls.

PROCEDURE:

1. Access Storage Section:

    a. Log in to the cloud platform console and navigate to the storage or buckets area.

2. Create a Bucket:

    a. Select "Create Bucket".
    b. Provide a unique name for the bucket.
    c. Choose a storage class (e.g., standard) and location.
    d. Set access permissions (e.g., private or public).
    e. Create the bucket.

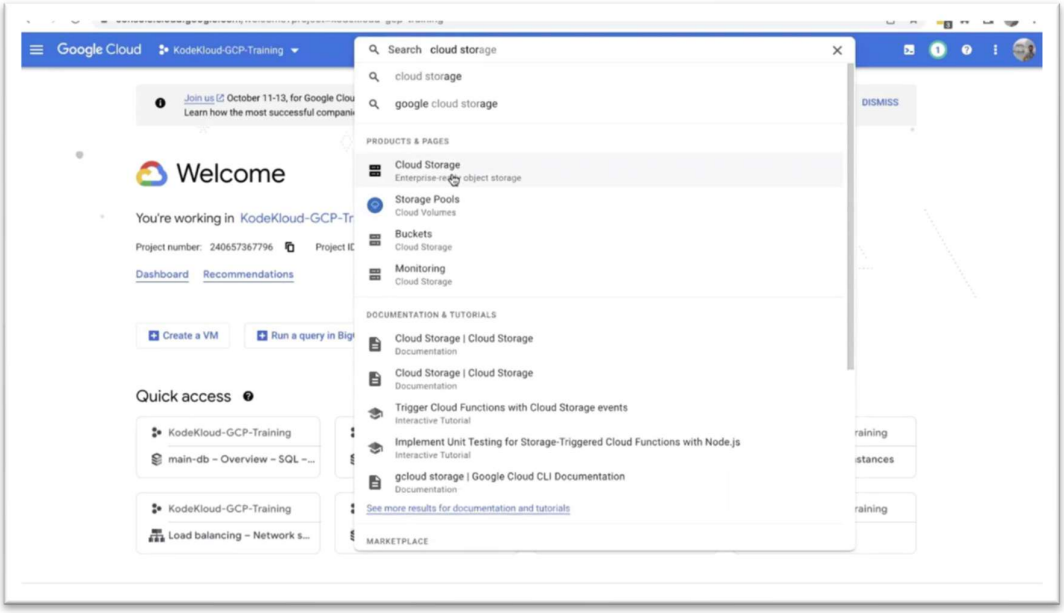
3. Upload a File:

    a. Open the bucket.
    b. Select "Upload" or drag and drop a file.
    c. Choose a local file and confirm upload.

4. Verify:

    a. Check the bucket contents to see the uploaded file.

RESULT: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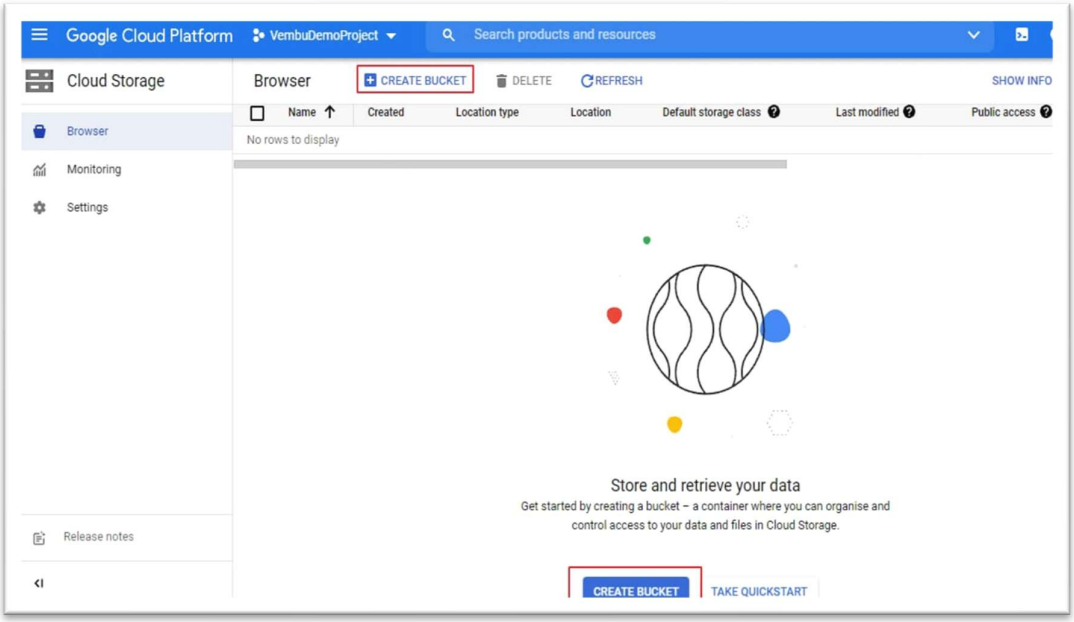
Load balancing – Network s...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Settings
Release notes
Browser
CREATE BUCKET
DELETE
REFRESH
SHOW INFO
Name
Created
Location type
Location
Default storage class
Last modified
Public access
No rows to display
Store and retrieve your data
Get started by creating a bucket – a container where you can organise and control access to your data and files in Cloud Storage.
CREATE BUCKET
TAKE QUICKSTART